# Haris Majeed Raja

(+92) 311-6652921 | harismajeed0501@gmail.com | linkedin.com/in/harismajeedraja | github.com/HarisMajeed05

## Summary

Started out building full-stack web apps and exploring core machine learning through internships in front-end development, ML, and Salesforce administration. Completed a final year project on multimodal deepfake detection using VLMs, and now working as an AI Engineer at MLBench, building computer vision pipelines. Looking to grow into a role where I can keep building production-grade AI systems, from detection models to full-stack applications, that solve real, practical problems.

## Education

**Information Technology University** — Lahore
*Bachelor of Science in Computer Science* — *Aug 2022 – Jun 2026*
*CGPA: 3.28/4.0*

## Experience

**AI Engineer** — Jun 2026 – Present
*MLBench Pvt Ltd*

- Worked on a field detection system for document forms, replacing a client's three-model pipeline (YOLO, FFDNet/FFDetr, and a geometry classifier) with a single fine-tuned **RF-DETR** model.
- Fine-tuned RF-DETR-Base on an annotated document dataset, reaching 0.9335 precision and 0.8496 recall against the client's existing three-model architecture at 0.9394 precision and 0.7875 recall, an F1 improvement from 0.8568 to 0.8896.
- Evaluated Tesseract and DocTR as OCR backends for the extraction pipeline, identifying checkbox glyph misreads and low-resolution footer text corruption in Tesseract output that DocTR resolved.
- Annotated client PDF documents for field geometry, type, and role to build the training set for the detection model, and logged experiment results to guide model selection.
- Contributed to a rice grain quality and variety analysis tool, training **DINOv2**-based classifiers across multiple image resolutions to grade grains as Full, Broken, Rejected, Weak, or Fatty and identify variety among Full grains.

**Teaching Assistant – Software Engineering & Artificial Intelligence** — Aug 2025 – Jun 2026
*Information Technology University*

- Assisted in teaching core Software Engineering and Artificial Intelligence concepts, including system design, version control, testing methodologies, search algorithms, and fundamentals of machine learning.
- Mentored students in developing scalable software and AI projects, guiding model selection, experiment design, and evaluation.
- Conducted code reviews, graded assignments and exams, and provided constructive feedback to strengthen student learning.
- Provided technical guidance on Python and ML frameworks including scikit-learn and PyTorch, and facilitated discussions linking course material with current AI research trends.

**Front-End Intern** — Jul 2025 – Aug 2025
*Certura*

- Built a responsive portfolio website using HTML, CSS, and JavaScript with animations and media query-based responsiveness.
- Developed a dynamic to-do list application and a weather dashboard using the OpenWeather API, with real-time data fetching and error handling.

**Machine Learning Intern** — Jun 2025 – Aug 2025
*Arch Technologies*

- Developed multiple ML models including spam detection using TF-IDF and Naive Bayes, housing price prediction using Linear Regression, and digit and iris classification.
- Applied preprocessing, feature extraction, and evaluation metrics including accuracy, $R^2$, MAE, and MSE using scikit-learn, NLTK, and matplotlib.

**Salesforce Admin Intern** — Jun 2024 – Sep 2024
*Incredibles*

- Completed in-depth Salesforce Administrator training via Trailhead, building strong practical expertise through hands-on projects.
- Developed a solid understanding of core Salesforce features including data modeling, reporting, dashboards, flows, and foundational Apex concepts.

## Projects

**Multimodal Deepfake Detection using VLMs (FYP)** Aug 2025 – Jun 2026
*Supervised by Dr. Waqas Sultani*

- Benchmarked state-of-the-art deepfake detectors (SBI, AltFreezing, Face Warping Detection) across FaceForensics++, Celeb-DF-v2, and HiDF, revealing that binary classifiers collapse to near-chance accuracy on high-quality deepfakes and have no mechanism to explain their predictions.
- Designed **ExDDV-Judge**, a novel three-stage detect, explain, judge pipeline that replaces 11B/7B-parameter VLMs with lightweight alternatives (BLIP-2 T5-XL, LLaVA-Mini), cutting model size by over 80% while running entirely on a single consumer GPU.
- Solved a critical real-frame misclassification bug in prior work by building a **Dual-DB routing system** with a fine-tuned CLIP encoder, taking real-frame accuracy from a broken 0% to 83%.
- Connected explanation generation with automated verification for the first time by pairing BLIP-2 T5-XL with **DeepfakeJudge**, producing faithfulness-scored explanations.
- Shipped a full-stack **Flask** web app delivering real-time, forensic-grade deepfake analysis, including per-frame labels, natural-language explanations, and trust scores, turning a research pipeline into a usable end-to-end tool.

**Legal AI Chatbot** | *GitHub Link* July 2026

- Built a full-stack RAG-based legal assistant using React, FastAPI, and MongoDB, with document-grounded Q&A powered by LangChain, FAISS, and the Groq API (Llama 3.3 70B).
- Implemented project-scoped retrieval so each client/case's uploaded PDFs are chunked, embedded, and queried in isolation, with every answer citing its source document.
- Added JWT-based authentication, persistent conversation memory, and real-time token streaming for a production-style chat experience.
- Designed the system to run GPU-free by offloading LLM inference to Groq's hosted API, keeping only lightweight CPU-based embeddings local.

**Footfall Counter** | *GitHub Link* June 2026

- Built a real-time people-counting system using YOLO/RT-DETR for detection and ByteTrack for persistent multi-object tracking across occlusion.
- Designed directional entry/exit counting via an arbitrary-angle virtual line, with an interactive tool to configure the line to match any camera setup.
- Optimized the full pipeline to run entirely on CPU, with configurable confidence, IoU, and resolution parameters for accuracy/speed trade-offs.

**C++ Compiler From Scratch** | *GitHub Link* Sep 2025 – Dec 2025

- Built a toy C++ compiler from scratch in C++ to explore core compiler design concepts.
- Implemented lexical analysis to tokenize input source code.
- Developed a recursive-descent parser for syntax analysis and basic semantic checks.
- Handled error detection and parsing of grammar defined in a custom specification.
- Explored the foundations of compiler pipelines, including future support areas such as code generation.

**Airbnb Clone (Extended)** | *GitHub Link* Oct 2024 – Dec 2024

- Developed a full-stack Airbnb-inspired web application using React, Node.js, and MongoDB.
- Implemented JWT-based authentication, role-based access control, and protected admin routes.
- Built features for property listing, booking management, and host-specific CRUD operations.

**Search Engine** | *GitHub Link* Oct 2023 – Dec 2023

- Built a custom search engine in C++ using an inverted index for efficient text retrieval.
- Implemented multi-condition query handling and result ranking.
- Deployed locally with a user-friendly command-line interface for querying.

## Technical Skills

**Programming Languages**: C++, Python, Java, C, SQL, HTML, CSS, JS
**Frameworks & Libraries**: React, Node.js, Express, FastAPI, Flask, MongoDB, scikit-learn, PyTorch, NumPy, pandas, Matplotlib, Gradio
**AI/ML Tools**: LangChain, FAISS, RAG Pipelines, LLM/VLM APIs (Groq, CLIP, BLIP-2, Qwen-VL), HuggingFace Transformers, Computer Vision (YOLO, RF-DETR, DINOv2, Cellpose-SAM, ByteTrack, OpenCV), OCR (Tesseract, DocTR)
**Platforms/Technologies**: Salesforce (Objects, Flows, Reports, Dashboards, Apex), JWT Authentication
**Tools & Environments**: Git, VS Code, Jupyter, Linux (Ubuntu), Logisim